INCH-POUND

MIL-C-55302/110C
7 June 1993
SUPERSEDING
MIL-C-55302/110F
31 October 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES PLUG, RIGHT ANGLE, 30 THROUGH 140 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS ( 100 SQ GRID)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302

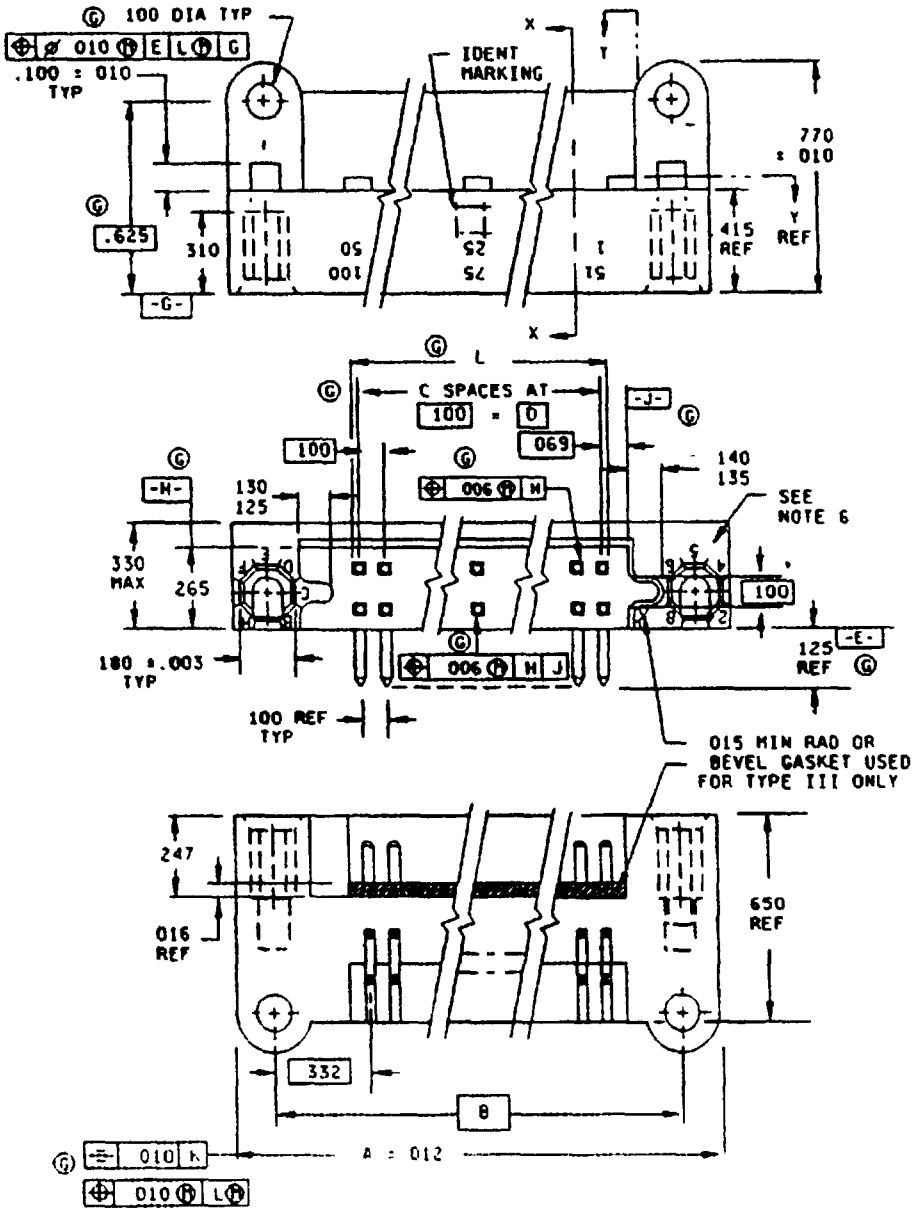



FIGURE 1 Connectors, plug ( 100 sq grid)

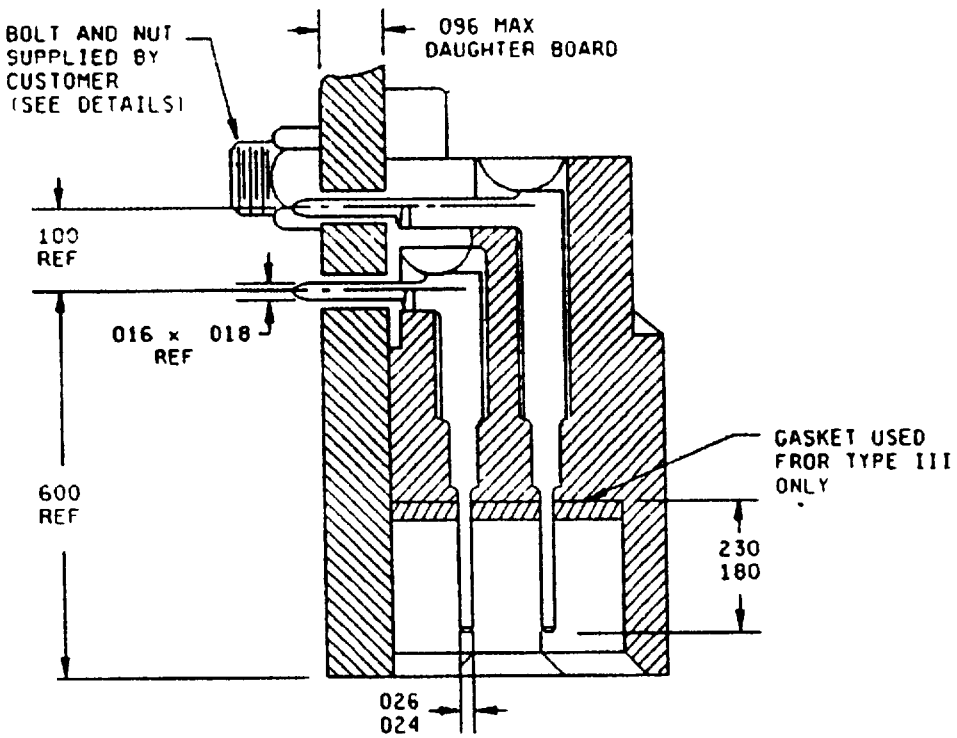
G denotes changes.



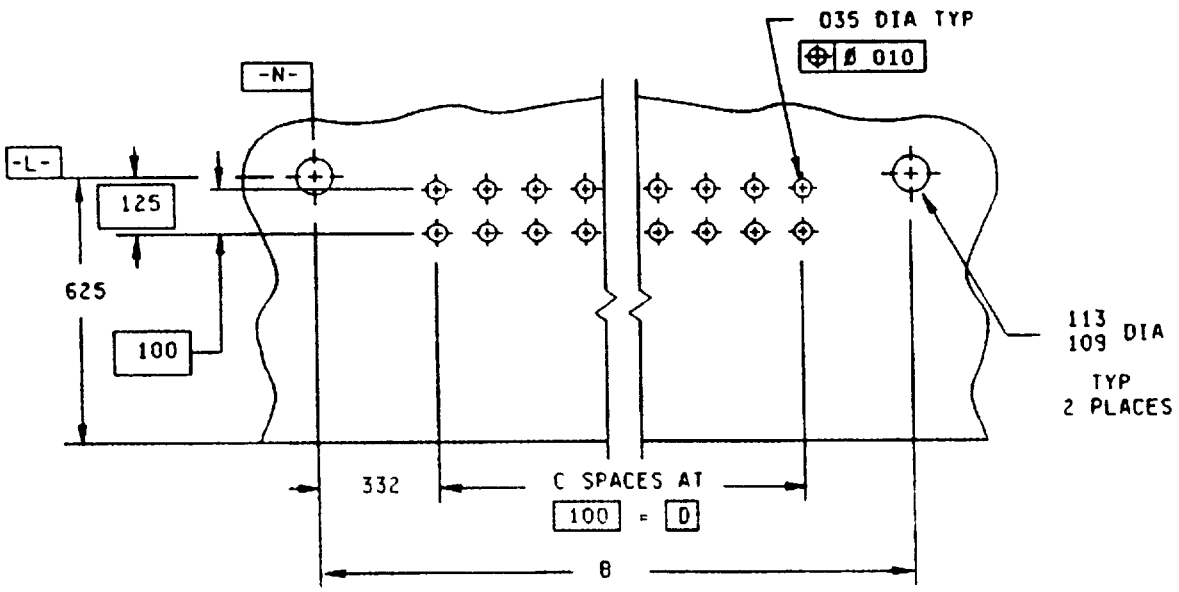
DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited.



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SECTION X-X

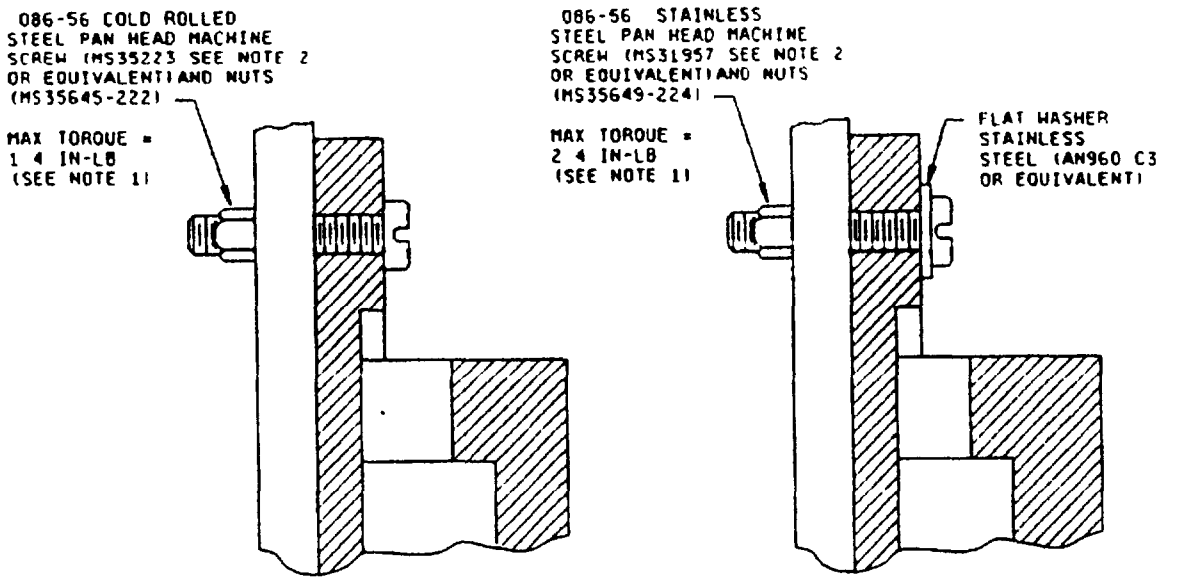



RECOMMENDED PC BOARD LAYOUT

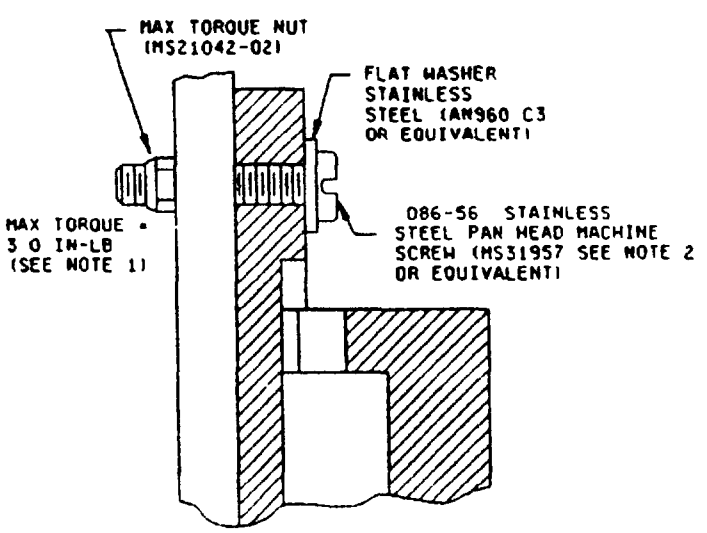
FIGURE 1 Connectors, plug ( 100 sq grid) - Continued

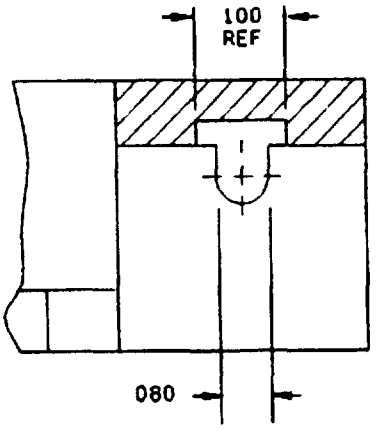


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MOUNTING RECOMENDATIONS




KEYING RIVET HOLE CONFIGURATION

SECTION Y-Y

NOTES
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 1. Connectors, plug, 100 sq grid) - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| 003 | 0 08 | 069 | 1 75 | 230 | 5 84 |
| 005 | 0 13 | 072 | 1 83 | 247 | 6 27 |
| 006 | 0 15 | 096 | 2 44 | 265 | 6 73 |
| 008 | 0 20 | 100 | 2 54 | 310 | 7 87 |
| 010 | 0 25 | 101 | 2 57 | 323 | 8 20 |
| 012 | 0 30 | 109 | 2 77 | 332 | 8 43 |
| 015 | 0 38 | 113 | 2 87 | 377 | 9 58 |
| 024 | 0 61 | 125 | 3 18 | 415 | 10 54 |
| 025 | 0 64 | 130 | 3 30 | 500 | 12 70 |
| 026 | 0 66 | 135 | 3 43 | 770 | 19 56 |
| 030 | 0 76 | 140 | 3 56 | | |
| 058 | 1 47 | 180 | 4 57 | | |

NOTES
1. Dimensions are in inches
2. Metric equivalent are given for formation only
3. Unless otherwise specified, tolerance is ±.005 (0 13 mm) on decimals and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/27
5. Numbers indicating every 5 cavities marked or molded on side Numbers indicating end cavities and line indicating every 5 cavities marked or molded on mating face
6. Key locations embossed on indicated surfaces

FIGURE 1 Connectors, plug ( 100 sq grid) - Continued



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TABLE I Dash numbers and dimensions for types II and III

| Dash number 2/ | Type number | Number of contacts | Dimensions 1/ | | | | Contact identification numbers |
|---|---|---|---|---|---|---|---|
| | | | A | B | C | B | |
| 10<br>19 | II<br>III | 30 | 2 355<br>(59 82) | 2.065<br>(52 45) | 14 | 1 400<br>(35 56) | 1, 16 |
| 11<br>20 | II<br>III | 40 | 2 855<br>(72 52) | 2.565<br>(65.15) | 19 | 1 900<br>(48.26) | 1, 21 |
| 12<br>21 | II<br>III | 50 | 3 355<br>(85.22) | 3 065<br>(77 85) | 24 | 2 400<br>(60 96) | 1, 26 |
| 13<br>22 | II<br>III | 60 | 3 855<br>(97.92) | 2 565<br>(90.55) | 29 | 2 900<br>(73 66) | 1, 31 |
| 14<br>23 | II<br>III | 70 | 4 355<br>(110 62) | 4 065<br>(103 25) | 34 | 3 400<br>(86 36) | 1, 36 |
| 15<br>24 | II<br>III | 80 | 4 855<br>(123 32) | 4 565<br>(115.95) | 39 | 3 900<br>(99 06) | 1, 41 |
| 16<br>25 | II<br>III | 90 | 5 355<br>(136 02) | 5 065<br>(128 65) | 44 | 4 400<br>(111 76<br>) | 1, 46 |
| 17<br>26 | II<br>III | 100 | 5 855<br>(148 72) | 5 565<br>(141.35 | 49 | 4 900<br>(124 46<br>) | 1, 51 |
| 18<br>27 | II<br>III | 110 | 6 355<br>(161 42) | 6 065<br>(154.05) | 54 | 5 400<br>(137 16<br>) | 1, 56 |
| 28<br>32 | II<br>III | 120 | 6 855<br>(174 12) | 6 565<br>(166 75) | 59 | 5 900<br>(149 86<br>) | 1, 61 |
| 29<br>33 | II<br>III | 130 | 7 355<br>(186 82) | 7.065<br>(179 45) | 64 | 6 400<br>(162 56<br>) | 1, 66 |
| 30<br>34 | II<br>III | 134 | 7 555<br>(191.90) | 7 265<br>(184 53) | 66 | 6 600<br>(167 64<br>) | 1, 68 |
| 31<br>35 | II<br>III | 140 | 7.855<br>(199.52) | 7.565<br>(192 15) | 69 | 6 900<br>(175 26<br>) | 1, 71 |

1/ Metric equivalents are in parentheses

2/ See table II for superseded part numbers

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REQUIREMENTS

Design and construction

Dimensions and configuration See figure 1 and table I

Material In accordance with MIL-C-55302

(G) Insulator body Insulator material shall be in accordance with MIL-C-55302 or MIL-M-24519, type GLCP-30F

(G) Contact material Insulator material shall be in accordance with MIL-C-55302 or beryllium copper alloy in accordance with ASTM B768

Gasket: Silicon rubber

Plating The contact plating for the engagement area, 150 inch minimum length, shall be gold over nickel in accordance with MIL-C-55302. The contact plating for the solder tail area, 160 inch minimum length, shall be tin lead over nickel in accordance wiyh MIL-C-55302 The remainder of the contact shall be nickel plated in accordance with MIL-C-55302

Contact identification See figure 1

Contact rating 3 0 amperes maximum per contact, 2 25 amperes continuous per contact at 75°F

Keying See MIL-C-55302/31, using the M55302/31-10 rivet

Mating and unmating The maximum mating force in pounds shall be the number of contacts multiplied by 25 and the withdrawal force in pounds shall be a minimum of 025 times the number of contacts and shall not exceed the measured insertion force

Contact resistance No individual contact pair shall have a resistance exceeding 020 ohm

Contact retention 3 pounds minimum

Dielectric withstanding voltage

Sea level 900 volts rms

High altitude 200 volts rms

Part number M55302/110- (and dash number from table I)

Superseded part numbers See table II

TABLE II Superseded part numbers

| Type I superseded part number M55302/110- | Type II superseding part number M55302/110- |
|---|---|
| 01 | 10 |
| 02 | 11 |
| 03 | 12 |
| 04 | 13 |
| 05 | 14 |
| 06 | 15 |
| 07 | 16 |
| 08 | 17 |
| 09 | 18 |



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Patent The Government has a royalty free license under the following patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government

US Patent number 3,404,367

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 17

Review activities
Army - AR, MI
Navy - AS, OS
Air Force - 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3907-03)



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